

your guard, rather than eliminating the menace of coronavirus you could end up increasing the threat. Ready to deep dive into the specifics?

Reduce viral load, reduce transmission

As we all now know, Covid-19 infection is most virulent indoors. The infection-causing virus that affects human cells is tiny (20 to 400nm) and spreads in aerosol form from the infected persons. The aerosol floats in the air for hours and infects the persons in the room who inhale a heavy load of virus. Thus, any solution that can eliminate or reduce the viral load in a closed space promises to be an effective solution.

UVC: The virus killer?

Since 1950s, UVC rays of 100-280nm wavelength have been recognised as the most powerful germicidal radiation that can damage the DNA of pathogens like virus, bacteria, and fungus. The UVC radiation has been shown to destroy the outer protein coating of SARS-coronavirus, which is effective in inactivating the Covid-19 causing virus.

However, for UVC radiation to inactivate viruses, direct (line-of-sight) exposure is a necessity. Plus, the dose of UVC radiation and time are important factors.

“UVGI technology for contactless, fast, and effective disinfection is now becoming an important and effective tool for sterilisation and decontamination of surfaces in hospitals, quickly, efficiently, at lower cost. Although the science of UVGI technology for pathogen sterilisation in hospitals has been known in the world and used since the last 70 years, its use is now becoming very popular for infection control, as antibiotics-resistant pathogens have rendered medicines ineffective. **Fortunately, there is no known pathogen that cannot be killed by a strong dose of UV germicidal irradiation; since UVC is not found in nature, pathogens have not developed resistance to it,**” shares Vineet K. Rohatgi, Binay Opto Electronics Private Limited, Kolkata.

Harness UVC to disinfect

UVGI has historically been used in unoccupied spaces for a variety of

What's UV? Is it not hazardous?

UV radiation is an electromagnetic wave energy that is also emitted by Sun. It has a shorter wavelength than visible light. The full spectrum of ultraviolet radiation is subdivided into UVA, UVB, and UVC. The UVC gets blocked by the atmosphere before reaching the earth's surface, but it can be created artificially with UV lamps.

UVA: 315-400nm wavelength

UVB: 280-315nm wavelength

UVC: 100-280nm wavelength

Out of the three, UVB is the most hazardous, as it can penetrate deeper into skin and eyes, and can lead to skin cancer and cataracts. UVA is not effective in penetrating pathogens.

Thankfully, UVC is both effective in penetrating pathogens (recognised by CDC of USA) and yet not so dangerous, as its long-term exposure causes skin burn or irritation only. If eyes are exposed then temporary blindness is induced, which one recovers from within a day or two.

Working of a UVGI System

The working of UVGI systems vary according to their purpose, but they commonly have high-power UVC tubes to sterilise the air passing through the coils. The high-energy photons, when absorbed by the DNA, disrupts them and makes the pathogen inactive. This has been the most effective way to inactivate germs like bacteria.

But viruses differ a bit with very thin protection layers. Viruses are the RNA/DNA code in a protective protein casing, unleashed inside a living cell to replicate the code and create more DNA copies to affect other cells. In viruses, sufficient UVC is needed to penetrate their shell and make irreplaceable damage to the RNA.

applications to disinfect food, air, water, etc but usually in unoccupied spaces only. However, recent innovations in UVGI have enabled the introduction of systems suitable to sanitise the surfaces of spaces like hospitals, offices, restaurants, fitness centres, and more in human presence. And they all are powered by UVC. Plus, UVC light has replaced chlorine in purifying water.

Surface decontamination through static or portable light disinfection, air decontamination through air handling units, HVAC disinfection, localised air sanitisers, upper-air disinfection and lower room luminaires, object decontamination through UVGI chamber, and water decontamination are primary applications of UVGI.

Sanitising surfaces and rooms

Sanitising of surfaces, commonly prone to spread of infections, can be done by UVC. This has led to portable handheld sanitising devices becoming popular in commercial markets. However, handheld devices will work

only when they get adequate time for exposure. Plus, as distance changes so does the strength of UVC, so the user must maintain the right distance and time to achieve decontamination. Else, a false sense of security may prevail, leading to potential disasters.

“At the time of purchase, users should check and verify the actual UVC intensity of the device. If the UVC output power is too low, the handheld device will be unable to eradicate pathogens, as time required to achieve the required dose will be impractically long,” explains Rohatgi. Plus, it would be prudent to wear eye protection goggles to avoid any damage to eyes.

Mobile carts with UVC lamps are commonly used for operation theatre disinfection in countries like the USA. These carts are robotically controlled or manually wheeled around the room. Many of them are now remotely controlled with the help of a handheld remote, mobile app, or power switches. But these are best used in the absence of humans. Some modern devices feature human-sensing capa-